1.	$\int (2x-5)(3x+2)(2x+5)\,\mathrm{d}x$ $(2x-5)(3x+2)(2x+5) = (6x^2-11x-10)(2x+5) = \ldots$ $= 12x^3+8x^2-75x-50$ $\int (2x-5)(3x+2)(2x+5)\,\mathrm{d}x = 3x^4+\frac{8}{3}x^3-\frac{75}{2}x^2-50x+c$	M1 A1 M1, A1ft, A1 **(5 marks)**

2.(a)	Attempts $A = \frac{1}{2}ab\sin C \Rightarrow 100 = \frac{1}{2}\times 25\times 15\sin BAC$	M1
	$\sin\theta° = \frac{8}{15}$	A1 **(2)**
(b)	$\left(BC^2\right) = 15^2 + 25^2 - 2\times 15\times 25\times \cos"\ \theta°"$ where $\theta° = \arcsin"\frac{8}{15}"$	M1
	$BC^2 = 15^2 + 25^2 - 2\times 15\times 25\times \cos\left(180 - \text{their '32.2'}\right) \Rightarrow BC = ...$	dM1
	$BC^2 = 1484.4... \Rightarrow BC = \text{awrt } 38.5\text{ cm}$ cso	A1 **(3)**
		(5 marks)

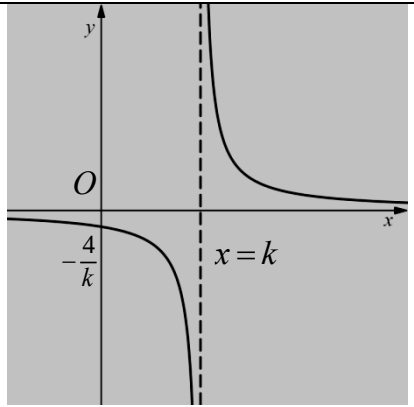
3.(a)	$y = \dfrac{5x^3 - 8}{2x^2} = \dfrac{5}{2}x - 4x^{-2}$ $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{5}{2} + 8x^{-3}$	M1 dM1 A1 A1 **(4)**
(b)	Substitutes $x = 2$ into their $\dfrac{\mathrm{d}y}{\mathrm{d}x} = "\dfrac{5}{2} + \dfrac{8}{2^3}" \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x} = "\dfrac{7}{2}"$ Uses their $\dfrac{7}{2}$ and (2, 4) $\Rightarrow y - 4 = "\dfrac{7}{2}"\left(x - 2\right)$ $\Rightarrow 7x - 2y - 6 = 0$	M1 M1 A1 **(3)** **(7 marks)**

4 (a)	$2\times 4^{x}-2^{x+3}=17\times 2^{x-1}-4$	
	Uses an index law and states or implies any of $4^{x}=p^{2},\quad 2^{x+3}=8p \quad \text{or} \quad 2^{x-1}=\frac{p}{2}$	B1
	Writes the given equation in terms of p $2\times 4^{x}-2^{x+3}=17\times 2^{x-1}-4\Rightarrow 2p^{2}-2^{3}\times p=\frac{17p}{2^{1}}-4$	M1
	Proceeds to $4p^{2}-33p+8=0$ via $2p^{2}-8p=\frac{17p}{2}-4$ * CSO	A1*
		(3)
(b)	$4p^{2}-33p+8=0\Rightarrow (4p-1)(p-8)=0\Rightarrow p=...,...$	M1
	Sets $2^{x}=\frac{1}{4},8\Rightarrow x=...$	M1
	$x=-2,3$	A1
		(3)
		(6 marks)

5 (a)	Attempts the gradient $\frac{\Delta y}{\Delta x}=\frac{9-6}{-2-10}=\left(-\frac{1}{4}\right)$	M1
	Uses the gradient and a point to form an eqn for l_1 $y-9=-\frac{1}{4}\left(x+2\right)$	dM1
	$y=-\frac{1}{4}x+\frac{17}{2}$	A1
		(3)
(b)	Eqn of l_2 is $y=4x$	B1ft
	Attempts to solve their $y=4x$ and their $y=-\frac{1}{4}x+\frac{17}{2}$ simultaneously	M1
	$R=(2,8)$	A1, A1
		(4)
(c)	Attempts $(OR)=\sqrt{"2"^2+"8"^2}$ or $(PQ)=\sqrt{3^2+12^2}$	M1
	Full attempt at area $OPQ=\frac{1}{2}\times\sqrt{3^2+12^2}\times\sqrt{2^2+8^2}$	dM1
	$=\frac{1}{2}\times3\sqrt{17}\times2\sqrt{17}=51$	A1
		(3)
		(10 marks)

6. (a)	$(540°, -5)$	B1, B1 **(2)**
(b) (i)	$(360°, 3)$	B1, B1 **(2)**
(ii)	$(180°, 5)$	B1, B1 **(2)**
		(6 marks)

7. (a)



Shape in quadrant One	M1
Fully correct shape and position	A1
C cuts the y-axis at $-\frac{4}{k}$	B1
C has a vertical asymptote at $x = k$	B1
	(4)

(b)

$$\frac{4}{x-k} = 9 - x \Rightarrow x^2 - (9+k)x + 9k + 4 = 0$$ M1, A1

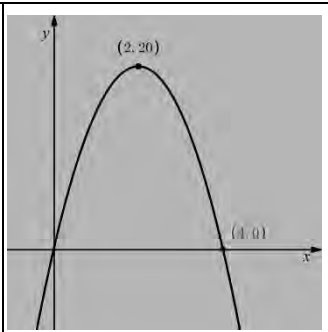
Uses $b^2 - 4ac < 0 \Rightarrow \left(9+k\right)^2 - 4\times 1\times\left(9k+4\right) < 0$ dM1

$k^2 - 18k + 65 < 0 \Rightarrow (k-13)(k-5) < 0 \Rightarrow 5 < k < 13$ ddM1, A1

(5)

(9 marks)

8 (a)	Attempts to use $S = r\theta \Rightarrow 9 = OD \times 0.8 \Rightarrow OD = 11.25$ or $\frac{45}{4}$	M1, A1
	$AO = \frac{5}{8} \times "11.25" = 7.03\,\text{m}$ *	A1*
		(3)
(b)	Attempts $A = 7.03 \times (2\pi - 0.8) = (38.55)$	M1
	Attempts $9 + 2 \times (11.25 - 7.03) + 7.03 \times \theta$	M1
	Perimeter = awrt 56.0 m	A1
		(3)
(c)	Attempts $\frac{1}{2} \times "11.25"^2 \times 0.8 = (50.625)$ OR $\frac{1}{2} \times 7.03^2 \times (2\pi - 0.8) = (135.4)$	M1
	Full method for area of platform = $\frac{1}{2} \times "11.25"^2 \times 0.8 + \frac{1}{2} \times 7.03^2 \times (2\pi - 0.8)$	M1
	=awrt 186 m^2	A1
		(3)
		(9 marks)

9 (a)	 Correct shape and position passing through (0, 0) Intersection at (4, 0)	B1 B1 **(2)**
(b)	Attempts form of equation. E.g. $y = Ax(x-4)$ or $y = 20 \pm C(x-2)^2$ Full attempt to find equation. E.g. $20 = A \times 2(2-4) \Rightarrow A = ...$ Or $0 = 20 + C(4-2)^2 \Rightarrow C = ...$ $y = -5x(x-4)$, $y = 20 - 5(x-2)^2$ o.e.	M1 dM1 A1 **(3)**
(c)	Sets $x\left(x^2 - 4\right) = "-5x(x-4)"$ $x^3 + 5x^2 - 24x = 0 \Rightarrow x\left(x^2 + 5x - 24\right) = 0$ $(x+8)(x-3) = 0 \Rightarrow x$ coordinate of P is -8 $P = \left(-8, -480\right)$	M1 dM1 ddM1, A1 A1 **(5)** **(10 marks)**

	$f'(x) = 4\sqrt{x^3} + \frac{k}{x^2} = 4x^{\frac{3}{2}} + kx^{-2}$	
10 (a)	$f''(x) = 6x^{\frac{1}{2}} - 2kx^{-3}$	M1, A1
	$f''(2) = 6\sqrt{2} - 2k \times \frac{1}{8} = 0 \Rightarrow k = 24\sqrt{2}$	dM1, A1
		(4)
(b)	$f'(x) = 4x^{\frac{3}{2}} + kx^{-2} \Rightarrow \quad f(x) = 4 \times \frac{2}{5}x^{\frac{5}{2}} - kx^{-1}\left(+c\right)$	M1, A1 ft
	Uses $P\left(2, 8\sqrt{2}\right) \Rightarrow 8\sqrt{2} = 4 \times \frac{2}{5} \times 2^{\frac{5}{2}} - \frac{k}{2} + c \Rightarrow c = p\sqrt{2}$	dM1
	$f(x) = \frac{8}{5}x^{\frac{5}{2}} - \frac{24\sqrt{2}}{x} + \frac{68}{5}\sqrt{2}$	A1
		(4)
		(8 marks)